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Temporal Fusion Transformer-Based Framework for Electric Vehicle Charging Demand Forecasting

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Abstract

Traditional forecasting approaches face challenges to accurately model the high temporal volatility of characteristics of electric vehicle (EV) charging demand, which is influenced by factors such as socioeconomic conditions, seasonal cycles, and regulatory incentives. In this paper, we present a Temporal Fusion Transformer (TFT) based predictive modeling framework for short-term, multi-horizon forecasting of station-level occupancy rates. The model combines static station metadata, known-future calendar/tariff signals, and observed-past local history with spatial neighbor aggregates to provide attention-based interpretability through variable selection and masked, interpretable multi-head attention. Evaluated on the ST-EVCDP dataset, TFT achieved very high accuracy when compared to other methods across nine stations at 5-minute resolution with an average mean absolute error of 0.0064 and high coefficient of determination (R^2) of 0.959, indicating outstanding predictive performance and resilience.

Keywords: Electric vehicle; Charging demand; Temporal Fusion Transformer; Forecasting; Multi-horizon

1. Introduction

The use of electric vehicles (EVs) has increased dramatically in recent years, changing road transportation and putting additional demand on electrical infrastructure. Global EV sales surpassed 17 million in 2024, which is accounting for more than 20% of all cars sold, and are predicted to exceed 20 million in 2025, with emerging economies rapidly growing and China driving development. Many countries, including the GCC countries, are now investing heavily in EV manufacturing and charging infrastructure, and are moving towards sustainable energy diversification and mobility electrification, e.g. Saudi Arabic through Lucid and CEER and the UAE through DEWA EV Green Charger and Tesla Supercharger programs.

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In order to guarantee dependable grids, effective charging infrastructure, and economical operation, this scale-up entails large, time-varying charging loads that must be anticipated[1]. Accurate EV charging demand projections support a wide range of operational and planning tasks, including distribution feeder capacity expansions, smart charging and pricing, station congestion management, and long-term infrastructure placement. According to recent engineering estimates, if electrification continues, a major portion of distribution feeders would need to be upgraded by the mid-2040's—costs that can be minimized by better forecasting and control. Bottom-up predictions for the United States also show significant, regionally varied charging loads until 2050, emphasizing the need for data-driven, location-aware approaches (Li and Jenn, 2024; Muratori and Yip, 2024).

Forecasting EV charging demand presents a challenge for traditional approaches for three reasons. First, charging is influenced by a variety of behaviors (such as trip timing, stay time, and state-of-charge) which result in high non-stationarity and intermittency at the station and feeder level. Second, exogenous forces like weather, traffic, prices, holidays, and local events influence demand over a wide range of time periods. Third, operators increasingly require calibrated multi-horizon forecasts (point and probabilistic) to support day-ahead scheduling and real-time control. Recent research confirms the predictive power of weather and traffic factors while emphasizing the significance of uncertainty quantification for practical application (Wang et al., 2025; Mystakidis et al., 2025; Amara-Ouali et al., 2025). Traditional statistical techniques (e.g., ARIMA and exponential smoothing) and recurrent neural networks (RNNs) such as LSTM/GRU have improved short-term EV load forecasts, but they struggle to combine diverse covariates, capture long-range dependencies, and remain interpretable for decision support purposes. Surveys of time-series transformers highlight their benefits for long-term forecasting and multivariate contexts. Deep learning models have shown promise in the EV domain, but many approaches continue to (i) under-utilize rich contextual signals (e.g., traffic flow, weather, calendar effects, prices), (ii) optimize for a single horizon rather than multi-horizon decisions, and (iii) provide limited interpretability or uncertainty estimates required for scheduling and pricing.

Transformer-based architectures have recently revolutionized time-series forecasting by modeling long-term temporal structure via attention and scaling to high-dimensional inputs. Within this family, the Temporal Fusion Transformer (TFT) developed an architecture designed for multi-horizon forecasting using static, known-future, and observed-past covariates Lim et al. (2021). TFT blends local sequence modeling with interpretable multi-head attention, variable selection networks to highlight significant drivers, and gating to suppress unnecessary components, resulting in excellent accuracy and actionable model diagnostics for operators. Building on the advantages of TFT, we present a TFT-based framework for EV charging demand forecasting that targets station-level occupancy rate $y \in [0, 1]$ using the ST-EVCDP dataset (Shenzhen). The framework categorizes feature engineering into three TFT-aligned groups: known-future reals (e.g., calendar terms), observed-past reals for capturing local dynamics and spatial context (e.g, target history y, raw occupancy, and neighbor aggregates), and static covariates (e.g., station identity and geography). It implements a leakage-free, time-based split with per-station normalization and reliable preprocessing for mixed units and scales. Furthermore, the framework trains a TFT for multi-horizon predictions and uses its built-in interpretability (via changeable selection weights and masked, interpretable attention). Empirically, our station-level evaluation shows low absolute error and high explained variance for short-term horizons, with consistent improvements from spatial context when neighbor coupling is substantial. The contributions of this paper can be summarized as follows: (i) A tailored TFT formulation for EV charging occupancy that systematically maps EV charging station data into known-future, observed-past, and static covariate groups; and (ii) An interpretable, multi-horizon evaluation protocol with station-level metrics and model diagnostics. The rest of the paper reviews the most relevant work (§2), describes the methodology and dataset (§3), and discusses the experiments and findings (§4). Finally, conclusions are summarized in (§5).

2. Related Work

Accurate forecasting of electric vehicle (EV) charging demand is essential for robust infrastructure planning, energy grid management, and effective policy-making. This section focuses on the study of approaches for forecasting electric-vehicle charging demand. We frame the literature across three linked areas: (1) classical time-series and econometric models, (2) deep sequential learners, and (3) attention-centric and Transformer-based approaches.

[1] https://www.iea.org/reports/global-ev-outlook-2025/executive-summary

ARIMA-family baselines remain common in EV charging studies due to their transparency and low data requirements. For instance, Amini et al. employed ARIMA to decouple conventional electrical load from EV parking-lot charging. They achieved a cleaner day-ahead scheduling. The model was interpretable and simple to fit; however it proved to be inadequate under nonstationary conditions and policy shocks (Amini et al., 2016). An Energies study (Kim and Kim, 2021) looked at large geographic areas to compare classical models (TBATS/ARIMA) with modern neural models. The study used charging data from different scales, from the national level down to a single station. It also included calendar and weather information. The classical models were good at explaining seasonal patterns. However, they were slow to adapt to sudden changes. They also could not link charging patterns between different locations . A recent Scientific Reports paper used a SARIMA model. It focused on power consumption at a single station. The model tracked seasonal signals well, but it did not react strongly enough to sudden spikes in demand. It also missed how different stations affected each other. These limitations highlight the need for more advanced models (Akshay et al., 2024).

Deep sequential learners, such as LSTM, GRU, and TCN, can achieve high accuracy when sufficient historical data and clean covariates are available. For instance, a World Electric Vehicle Journal paper (Tian et al., 2025) used particle-swarm optimization (PSO) to tune an LSTM for charging-pile loads. This model consistently outperformed standard LSTM/GRU variants across seasons. It demonstrated strong sequence modeling; however, the approach was data-intensive and limited to single-site analysis. Zhu et al. studied minute-level charging data from Shenzhen (Zhu et al., 2019). They found LSTM models beat simpler baselines for very short-term forecasts. This shows LSTMs are valuable for handling intermittent data. Another study by Zhang et al. wanted to use multi-scale patterns and external data, like weather (Zhang et al., 2022). They combined multi-channel CNNs with TCN blocks. However, these models struggled to use long-range patterns or many different types of inputs. Besides, they are often optimized for only one-step-ahead forecasts. Operators, however, need multi-horizon forecasts to make real-world decisions.

Transformer-based models can be more accurate for long-term forecasts compared to ARIMA or LSTM models. This is because they can analyze the entire data history at once (a ”global receptive field”). Recent EV studies use attention mechanism and transformer-based architectures. Koohfar et al. used transformer-based architecture on station data from Boulder, Colorado (Koohfar et al., 2023). They included weather and calendar data. Another study by Hussain et al. combined an LSTM with a Transformer (Hussain et al., 2025). They tested it on the open ACN (Caltech/JPL) datasets. This model improved accuracy (MAE/MSE) over very long periods (30 to 240 days). This result shows that attention models are robust for long-horizon forecasting.

3. Methodology

The EV charging demand is typically modeled as a time series. Our proposed methodology is based on Temporal Fusion Transformer (TFT) which provides as robust and interpretable framework for multi-horizon forecasting (Lim et al., 2021). Additionally, TFT can combine past observable data, known future data, and covariates that do not change over time. A conceptual overview of the problem and proposed methodology is illustrated in Fig. 1. In the following subsection, we describe the data preparation and forecasting methodology.

3.1. Data preparation

We employ the Shenzhen ST-EVCDP dataset (Kuang et al., 2024) collected at five-minute resolution from 247 charging stations in Shenzhen, China during June 19 to July 18, 2022 (30 days), with a total of 18,061 public charging piles. For faster raining and efficient resource usage while at the same time maintaining the accuracy we adopt only 9 station data. Raw files (volume, occupancy (rate) as the target, duration, price, information, adjacency, distance, and time) are aligned on a common timeline and merged into a single long-format spatio-temporal panel indexed by time and zone. Numerical series are coerced to numeric, reindexed on the five-minute grid, and cleaned per zone via linear interpolation with median backfilling for residual gaps.

To encode temporal, spatial, and contextual structure, we engineer calendar features (hour of day, day of week, weekend indicator) and cyclic encodings of time of day using sine and cosine transforms; price and calendar variables are treated as known-future covariates, whereas recent histories of demand (volume), occupancy, and duration are treated as observed-past covariates. Spatial influence is incorporated by row-normalizing the adjacency matrix to

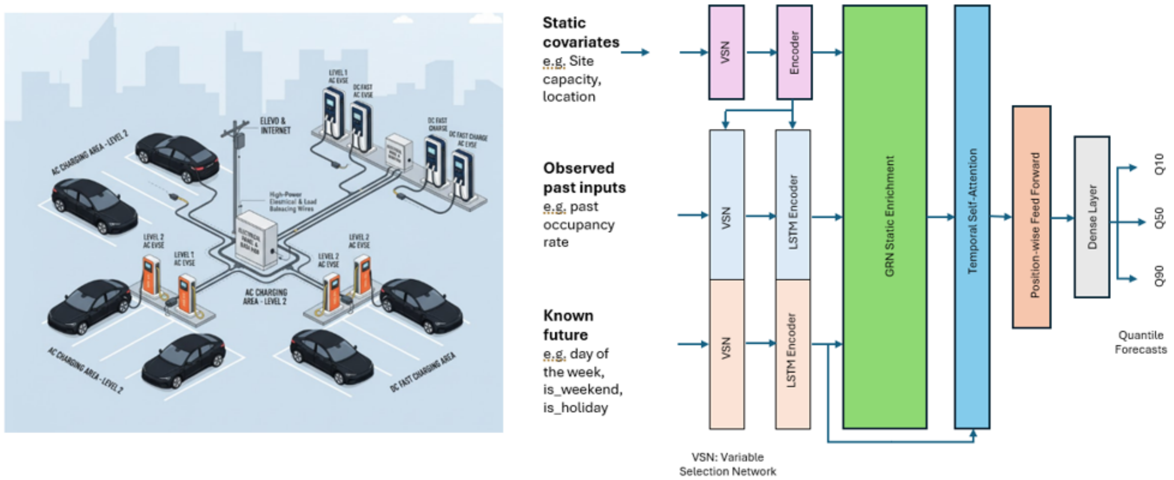

Fig. 1. Conceptual illustration of one station, at the left, and a high-level overview of the proposed forecasting methodology, at the right.

obtain a weighting matrix W and by deriving mean neighbor distances from the distance matrix; neighbor-averaged dynamics (nbr_volume, nbr_occupancy, nbr_duration, nbr_price) are computed by multiplying zone-wise wide matrices with W. Static per-zone features (pile capacity, latitude, longitude, graph degree, and neighbor count, mean neighbor distance, and the two policy flags) are treated as time-invariant inputs alongside the dynamic covariates. This feature engineering process is aligned with the feature groups required by TFT.

3.2. Temporal Fusion Transformer for EV Charging Demand Forecasting

TFT is selected for its ability to fuse heterogeneous inputs through gated residual networks and variable-selection layers, to capture temporal dependencies with an LSTM-based encoder–decoder, and to provide interpretability via attention. The technical details of each component are explained below. TFT combines (i) gating for robust deep stacks, (ii) variable selection for time-local interpretability, (iii) a static covariate encoder to condition all modules, (iv) encoder-decoder sequence modeling fused with interpretable multi-head attention, and (v) quantile outputs for calibrated multi-horizon forecasts (Lee et al., 2025). TFT uses static variables, time-varying inputs, and self-attention to generate multi-horizon forecasts. It combines future known drivers with encoded history in a single predictive design. The Gated Residual Network (GRN) in TFT utilizes a gated residual network with a primary vector p and a context vector c; gating is implemented by a gated linear unit (GLU), and activation is ELU. This enables context-aware nonlinear transforms while filtering out irrelevant data.

$$\mathrm{GRN}(\mathbf{z};\mathbf{c}) = \mathrm{LN}\Big(\mathbf{z}_{\mathrm{skip}} + \underbrace{\mathrm{GLU}\big(W_2\,\phi(W_1\mathbf{z} + V\mathbf{c} + \mathbf{b}_1) + \mathbf{b}_2\big)}_{\text{gate (GLU), activation (ELU)}}\Big) \tag{1}$$

Moreover, Variable Selection Networks (VSN) are used to pick variables for both static and time-dependent covariates, maintaining fundamentals while removing unnecessary components. The selection is time-local and conditional on the encoder's static context. Furthermore, encoder-decoder as a sequence-to-sequence backbone captures local temporal relationships; the encoder summarizes history, while the decoder rolls forecasts forward using known-future drivers.

$$\mathbf{z}_t = \sum_{v\in\mathcal{V}} \underbrace{\mathrm{softmax}_v\big(\mathbf{w}^\top \mathrm{GRN}(\mathbf{e}_{t,v};\mathbf{c})\big)}_{\alpha_{t,v}}\,\mathbf{e}_{t,v}, \tag{2}$$

$$(\mathbf{h}_t^{\text{enc}}, \mathbf{m}_t^{\text{enc}}) = \text{LSTM}_{\text{enc}}(\mathbf{z}_t^{(o)}, \mathbf{h}_{t-1}^{\text{enc}}, \mathbf{m}_{t-1}^{\text{enc}}), (\mathbf{h}_t^{\text{dec}}, \mathbf{m}_t^{\text{dec}}) = \text{LSTM}_{\text{dec}}(\mathbf{z}_t^{(k)}, \mathbf{h}_{t-1}^{\text{dec}}, \mathbf{m}_{t-1}^{\text{dec}}) \tag{3}$$

A static enrichment layer in the temporal fusion decoder with interpretable attention combines time-series features into a matrix and applies interpretable multi-head attention to all time steps, with weights shared throughout the layer. A self-attention layer and a position-wise feed-forward with additional gating are also provided to aid training. Finally, the quantile output head is used to generate predictions with a linear head per quantile q, resulting in calibrated multi-horizon outputs from the temporal fusion decoder.

$$\hat{y}_{t_0+h}^{(q)} = W_o^{(q)} \mathbf{u}_{t_0+h} + b_o^{(q)}, \qquad q \in \mathcal{Q},\ h = 1, \ldots, H \tag{4}$$

The static encoder provides context for temporal variable selection, local temporal feature extraction, and temporal feature enrichment applied before attention, ensuring that station-level attributes influence both selection and fusion phases. This design improves stability and selective information flow; VSNs allow for time-step interpretation; the seq2seq core captures short- and mid-range locality; interpretable multi-head attention captures long-range dependencies with shared value structure; and quantile heads enable operationally useful uncertainty. In our configuration, the encoder consumes a three-hour history (36 five-minute steps) and the decoder forecasts a one-hour horizon (12 steps) at each reference time.

4. Evaluation

4.1. Experimental setup and evaluation metrics

The performance of different models is evaluated using the standard regression metrics, including Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE), and Coefficient of Determination (R^2). These metrics are mathematically defined as follows:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|, \quad \text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}, \quad \text{MAPE} = \frac{100}{n}\sum_{i=1}^{n} \left|\frac{y_i - \hat{y}_i}{y_i}\right|, \quad R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{5}$$

where n is the number of samples, y_i is the actual value, $\hat{y}_i$ is the predicted value, and $\bar{y}$ is the mean value. While MAE works well for interpretability and balanced evaluation of forecasting performance, it treats large and small errors linearly. In contrast, RMSE penalizes larger errors more when large deviations or outliers are critical. MAPE expresses the error as a percentage, making it easier to compare the performance across different datasets of different scales. Finally, R^2 quantifies the proportion of variance explained by the model, which is very useful to compare different models on the same dataset.

4.2. Experiments and results

The experimental results for nine individual stations are shown in Table 1. From the table, it is observed that the TFT performs well in the short term for projecting occupancy rates at nine typical stations. Averaged over stations, the model achieves $MAE = 0.0064$, $RMSE = 0.0098$, $MAPE = 4.24\%$, and $R^2 = 0.959$. Because the target rate is limited to $[0, 1]$, these values amount to an average absolute error of around 0.64 percentage points of occupancy at a 5-minute resolution, suggesting close tracking of intra-day demand fluctuations and strong multi-horizon performance.

Table 1. Per-station TFT performance (target: occupancy rate). MAPE shown in %.

station_id	**n_samples**	**MAE**	**RMSE**	**MAPE**	$\mathbf{R^2}$
108	1681	0.0071	0.0120	1.7477	0.9817
109	1681	0.0050	0.0093	4.2653	0.9780
107	1681	0.0037	0.0063	1.8150	0.9742
102	1681	0.0094	0.0163	1.6738	0.9702
111	1681	0.0038	0.0056	2.0277	0.9661
123	1681	0.0070	0.0109	4.1211	0.9580
110	1681	0.0046	0.0062	3.7662	0.9438
105	1681	0.0073	0.0092	4.4833	0.9383
115	1681	0.0098	0.0125	14.2505	0.9194
Avg		**0.0064**	**0.0098**	**4.2390**	**0.9588**

It is further observed that performance varies among stations, indicating true station-level variation. Stations with smoother, capacity-driven cycles and stronger spatial coupling to neighbors tend to yield smaller errors (e.g., station 107 with $MAE = 0.0037$ and $R^2 = 0.974$), whereas stations exhibiting irregular or event-driven demand, weaker neighborhood signals, or potentially mis-specified inputs show larger deviations (e.g., station 115 with $MAPE = 14.25\%$ despite moderate absolute errors). At low-load regimes, small denominators magnify percentage errors even while *MAE*/*RMSE* remain modest, causing *MAPE* to inflate. This is a recognized phenomenon when the target approaches zero. Neighborhood aggregates (e.g., `nbr_occupancy`, `nbr_volume`, `nbr_duration`, `nbr_price`) provide a stabilizing spatial context that improves stations with sparse or noisy signals. Static station metadata (capacity, time-pricing indicators, spatial degree and neighbor counts) is incorporated into site-specific dynamics via static enrichment, tailored gating, variable selection, and LSTM initialization.

The model's interpretability mechanisms support these qualitative findings. Variable Selection Networks identify which covariates are most important at each time step. In practice, calendar terms and salient neighbor signals often dominate, with price importance increasing with scheduled tariffs. The masked, interpretable multi-head attention focuses probability mass on important historical patterns—often previous days at similar times or post-event rebound periods—which explains how the decoder uses encoded history for each horizon step. These diagnostics work together to provide decision support in addition to raw error metrics. Scale-aware metrics, such as *MAE* and *RMSE*, are appropriate at a bounded rate. However, *MAPE* should be interpreted cautiously under low occupancy. For operational readiness, horizon-wise error profiles and uncertainty calibration are as important as averages. Errors typically grow with horizon, and quantile outputs should be assessed by coverage (e.g., 50%/90%) to ensure forecasts are not only accurate but also reliably calibrated for use in congestion mitigation, dynamic pricing, and staffing. The forecasted vs. actual occupancy of the nines stations are visualized in Fig. 2.

Overall, low absolute error, high explained variance, and consistent interpretability support TFT as a practical, deployable framework for multi-horizon EV charging occupancy forecasting. Additionally, it also provides clear avenues for robustness checks (e.g., ablations on neighbor signals and static features) and extension to broader seasonal windows.

4.3. Comparison with the Baselines

We benchmarked TFT against three standard baselines: a Naive (last value) model that projects the most recent observation forward across all horizons (a strong persistence reference), a Moving Average (12) that uses the trailing mean of the last 12 time steps to smooth noise and capture slow-moving trends, and ARIMA/SARIMAX (AutoRegressive Integrated Moving Average / Seasonal ARIMA with exogenous regressors), a classical linear time-series approach that models autoregressive dynamics, differencing for non-stationarity, moving-average shocks, and seasonal/exogenous effects.

The scores in Table 2 compare the proposed TFT to the bseline models. It is observed that TFT achieves the best accuracy across the three error metrics: MAE = 0.0064, RMSE = 0.0098, and MAPE = 4.239%, surpassing the

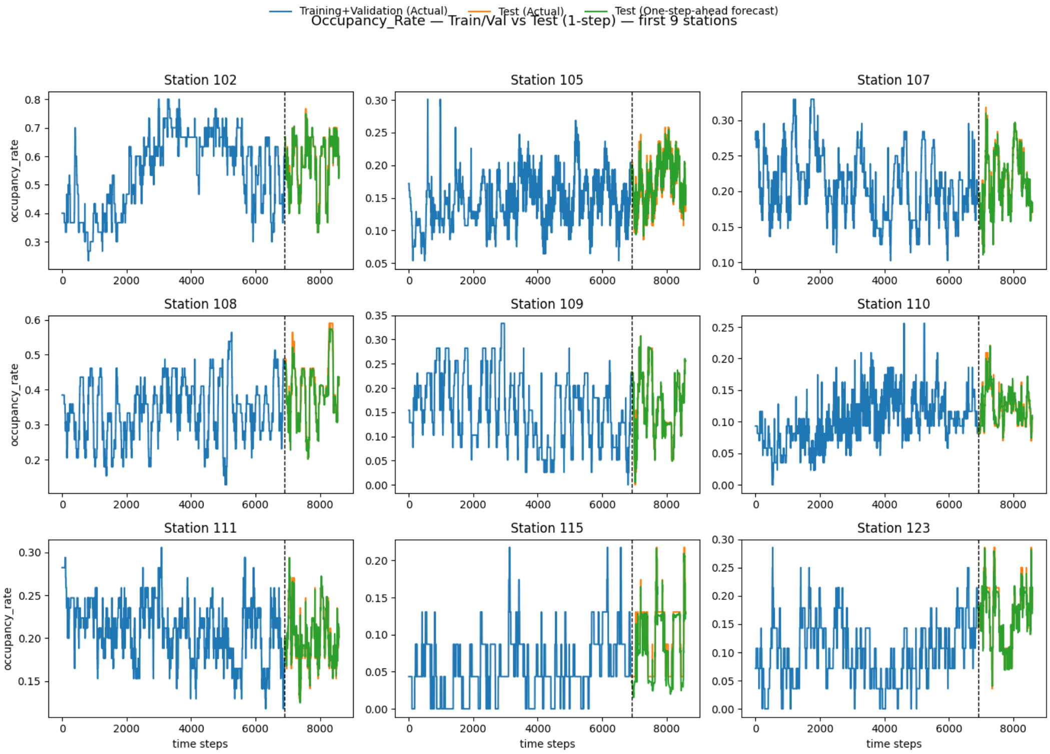

Fig. 2. One-hour ahead forecasted vs. actual occupancy rate

Table 2. Comparison of the proposed method with the baselines (target: occupancy rate). MAPE shown in %. Best values per metric in **bold**.

Model	MAE	RMSE	MAPE	R^2	Overall Rank
TFT	**0.0064**	**0.0098**	**4.2390**	0.9588	1
Naive	0.0175	0.0307	9.5070	**0.9613**	2
MovingAvg(12)	0.0213	0.0341	11.7281	0.9522	3
arima/sarimax	0.0480	0.0729	35.5730	0.7821	4

next-best baseline by a considerable margin. Compared to Naive, TFT reduces MAE by 63.4%, RMSE by 68.1%, and MAPE by 55.4%. Improvements against MovingAvg(12) are equally strong (MAE 70.0%, RMSE 71.3%, MAPE 63.9%), whereas ARIMA/SARIMAX trail significantly (TFT increases MAE/RMSE/MAPE by 86–88%). These benefits demonstrate TFT's capacity to combine diverse covariates (calendar features, scheduled pricing when available, and neighbor aggregates) with static enrichment, variable selection, and temporal attention, capturing both recurring patterns and context-driven variations.

The Naive baseline has a slightly higher R^2 (0.9613) than TFT (0.9588), despite significantly more pointwise errors. This behavior is typical when the goal is highly autocorrelated and bounded: persistence might imitate overall variance around the mean (inflating R^2) while missing turning points that dominate MAE/RMSE/MAPE. To estimate occupancy rates, scale-aware accuracy metrics (MAE/RMSE) and percentage error (MAPE) are more diagnostic than R^2 alone. Furthermore, the moving average baseline performs poorly due to smoothing lag, which attenuates peaks and valleys; ARIMA/SARIMAX is hampered by non-stationarities and a limited ability to use external and spatial information when compared to TFT.

Overall, TFT's constant dominance in error metrics and built-in interpretability (varying priority and attention patterns) justifies its top rank in Table 2. In practice, these accuracy advantages translate into tighter management of congestion risk and improved scheduling/pricing decisions. However, the tiny R^2 edge of Naive highlights the significance of adopting a balanced set of metrics and horizon-wise diagnostics for constrained, autocorrelated objectives. This is also attributed to the high variance in the data but does not directly capture the forecast accuracy.

5. Conclusion

In this paper, we proposed a TFT-based methodology for multi-horizon forecasting of EV charging occupancy rate on the ST-EVCDP dataset, combining static station metadata, known-future calendar/tariff signals, and observed-past local and neighbor aggregates. Across nine stations at 5-minute resolution, the model achieves average MAE = 0.0064, RMSE = 0.0098, MAPE = 4.24%, and R^2 = 0.959, yielding very low absolute errors while providing built-in interpretability via variable selection and temporal attention for energy operators to support in decision making. However, the tariff inputs should be known ahead of time, and percentage metrics should be read with caution at low occupancy. Overall, TFT's gating, selection, and attention constitute a compact, accurate, and transparent forecasting core. Future extensions will include broader validation across more stations and seasons and explicit spatial learning (e.g., graph-based attention) to supplement neighbor aggregates. Despite these limitations, the research supports TFT as a solid and understandable foundation for energy operators in EV charging demand forecasts.

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